

Practice Quiz: Communication

Part A: Multiple Choice

1. Mirror neurons fire: A) Only during the execution of actions B) Both during action execution and during observation of the same action performed by another C) Only during language comprehension D) Only in the visual cortex
2. In Active Inference, mirror neuron activity represents: A) Simple motor imitation B) Motor simulation — the observer's generative model predicting the proprioceptive consequences of observed actions C) Emotional contagion D) Random motor noise
3. Interpersonal neural synchrony is best described as: A) Two brains generating identical neural activity simultaneously B) Correlated oscillatory activity between interacting brains, reflecting alignment of generative models C) Physical proximity between people D) Emotional similarity between conversational partners
4. The mentalizing network (TPJ, mPFC, pSTS, temporal poles) is primarily involved in: A) Visual perception B) Representing others' beliefs, desires, and intentions C) Motor control D) Auditory processing
5. The false belief task measures: A) Memory for object locations B) The ability to attribute a belief to another person that differs from reality — a core Theory of Mind capacity C) Visual tracking ability D) Language comprehension
6. In schizophrenia, paranoid ideation may result from: A) Absent Theory of Mind B) Overactive mentalizing — excessive attribution of intentions (often threatening) to others C) Damage to the visual system D) Complete loss of mirror neuron function
7. Successful speaker-listener communication in hyperscanning studies correlates with: A) Identical brain anatomy B) Tighter temporal coupling of neural activity between speaker and listener C) The listener's ability to speak the same dialect D) Higher intelligence in both participants

Part B: Short Answer

1. Explain how Active Inference's concept of "generalized synchrony" accounts for the neural coupling observed in hyperscanning studies. What must align between two agents for communication to succeed?
2. A child under age 4 fails the false belief task. Analyze this in Active Inference terms: what level of the generative model hierarchy is not yet developed?
3. An adult with autism understands literal language but misses sarcasm and irony. Design a study to test whether this reflects reduced precision on mentalizing predictions or a fundamentally different social inference strategy.